
Smoothed Analysis of Simplex via Circuits

The Klee–Minty cube of Lecture 2 shows that the simplex method can be forced to take exponentially many pivots. Such worst-case instances, however, are *brittle*: a slight perturbation of the instance data destroys the long simplex path. This observation was turned into a theory by Spielman and Teng [ST04]: in *smoothed analysis*, an adversary picks an arbitrary instance, a small random perturbation is added, and one bounds the *expected* running time on the perturbed instance. A polynomial smoothed bound shows that hard instances are isolated points in instance space, which are eliminated by small perturbations. Spielman and Teng proved that the shadow vertex simplex method solves LPs whose constraint data is perturbed by Gaussian noise of standard deviation σ in expected time polynomial in the dimensions and $1/\sigma$ — a landmark explanation of the practical behavior of the simplex method (with by-now much sharper bounds, see [DH18]).

Rather than the general (and technical) Gaussian analysis, this lecture carries out two complete smoothed analyses in settings where only the *objective* is perturbed. The first, following the presentation in [DH21], concerns the *successive shortest path* (SSP) algorithm for minimum-cost flow (Section 2); the result is due to Brunsch, Cornelissen, Manthey, Röglin and Rösjner [BCM⁺15]. The second (Section 4) extends the analysis from flows to general LPs whose constraint matrix has bounded *circuit imbalance*. The thread joining the two — and the namesake of this lecture — is the notion of a *circuit* of the constraint matrix: circuits simultaneously describe the edge directions walked by the simplex method, the augmenting cycles and paths of network flows, and the condition measure κ that governs the general case. The circuit toolkit — conformal decompositions and the imbalance measure — is developed in Section 3, between the two analyses; it will serve again in the interior point lectures.

1 Preliminaries

1.1 The toolkit from Lecture 2

We briefly recall the required material from Lecture 2, to which we refer for details. We work with LPs in standard form,

$$\min c^\top x \text{ s.t. } Ax = b, \quad x \geq 0,$$

where $A \in \mathbb{R}^{m \times n}$ has full row rank $m \leq n$. A *basis* is a set $B \subseteq [n]$ with $|B| = m$ and A_B (the corresponding columns) invertible; the associated *basic feasible solution* (BFS), when $x_B := A_B^{-1}b \geq 0$, is the vertex of the feasible region with support in B and $x_N = 0$ on $N := [n] \setminus B$. From a BFS with basis B , the simplex method moves along *pivot directions*: for an entering index $i \notin B$, the direction $w^{B,i}$ given by

$$w_i^{B,i} = 1, \quad w_B^{B,i} = -A_B^{-1}A_i, \quad w_j^{B,i} = 0 \text{ otherwise,}$$

which generate the edges of the feasible region incident to the vertex.

Our runs of the simplex method will follow the *shadow vertex* (parametric objective) rule of Lecture 2, in an orientation chosen to align with the SSP algorithm: we consider the shadow path induced by a *starting objective* c , uniquely minimized at the starting vertex v_0 , and a *target objective* $-d$ — the walk ends at a maximizer of d . We call this the *shadow path for the pair* (c, d) . The rule maintains a vertex minimizing the interpolated objective

$$c_\lambda := (1 - \lambda)c + \lambda(-d) = (1 - \lambda)c - \lambda d$$

as λ sweeps from 0 to 1. Its key property is *locality*: whether a vertex is visited depends only on the data — namely, on whether the vertex minimizes some intermediate objective c_λ — and not on the history of the walk. Geometrically, project by

$$\pi(x) := (d^\top x, c^\top x),$$

so that the target d is the *horizontal* direction and the cost c the *vertical* one. The visited vertices are precisely the vertices of the feasible region projecting onto the lower boundary of the two-dimensional *shadow* $\pi(P)$ to the right of $\pi(v_0)$ — the trade-off curve recording the least c -cost at each attainable d -value — traversed rightward from the c -minimal vertex to the d -maximal one. In this lecture the starting objective will come for free: SSP starts at the zero flow, the unique minimizer of the (positive) flow cost, and pushes toward maximum flow value.

We also recall the reduction of variable bounds to standard form.

REMARK 1 (VARIABLE BOUNDS) Capacitated programs, $\min c^\top x, Ax = b, \mathbf{0} \leq x \leq u$, reduce to standard form by pairing each variable with a complementary slack variable: $x_j + \bar{x}_j = u_j, x, \bar{x} \geq \mathbf{0}$. The kernel of the enlarged matrix is $\{(w, -w) : w \in \ker(\mathbf{A})\}$ — a mirrored copy of $\ker(\mathbf{A})$ — so every kernel-based quantity (edge directions, circuits, and the circuit imbalance measure of Section 3) carries over unchanged, while each bound $x_j \leq u_j$ becomes the nonnegativity of the complementary variable $\bar{x}_j$. In this lecture, the complement of a flow variable is precisely its reverse residual arc.

1.2 The local step: a program over the tangent cone

The pivot step of the shadow vertex rule has a clean *local* description, which will let us recognize familiar combinatorial algorithms as shadow walks. Recall from Lecture 2 the *tangent cone* of P at a vertex v ,

$$T_P(v) := \{y : v + \varepsilon y \in P \text{ for some } \varepsilon > 0\},$$

the cone of feasible directions at v , generated by the directions of the edges of P incident to v . In standard form, $T_P(v) = \{y \in \ker(\mathbf{A}) : y_j \geq 0 \ \forall j \text{ with } v_j = 0\}$: equality constraints must be preserved, tight nonnegativity constraints must not be violated, and the generators are the pivot directions $w^{B,i}$ above. Local optimality is optimality over the tangent cone: v minimizes an objective q over P if and only if $q^\top y \geq 0$ for all $y \in T_P(v)$ (Lecture 2).

LEMMA 1 (THE TANGENT CONE PROGRAM) Let v be a vertex on the shadow path for (c, d) other than its endpoint, and suppose v minimizes c_λ exactly for $\lambda \in [\lambda^-, \lambda^+] \subseteq [0, 1]$. Then the edge direction followed by the path out of v is the optimal solution y^* of the tangent cone program

$$\min c^\top y \text{ s.t. } y \in T_P(v), \quad d^\top y = 1, \tag{1}$$

unique under our standing non-degeneracy assumptions, and the optimal value — the slope of the outgoing edge in the shadow plane — equals

$$c^\top y^* = \frac{\lambda^+}{1 - \lambda^+}.$$

PROOF: Since v minimizes c_{λ^+} , local optimality gives $c_{\lambda^+}^\top y = (1 - \lambda^+) c^\top y - \lambda^+ d^\top y \geq 0$ for every $y \in T_P(v)$; so every feasible y of (1) has $c^\top y \geq \lambda^+ / (1 - \lambda^+)$. Now let y^* span the outgoing edge.

For λ slightly above λ^+ the next vertex $v + ty^*$ ($t > 0$) is the minimizer of c_λ , so $c_\lambda^\top y^* \leq 0$, while $c_{\lambda^+}^\top y^* = 0$ (at λ^+ both endpoints of the edge are optimal, so the edge lies on the minimizing face of c_{λ^+}). Feeding $c_{\lambda^+}^\top y^* = 0$ into the strict inequality $c_\lambda^\top y^* < 0$ for $\lambda > \lambda^+$ forces $d^\top y^* > 0$; normalizing $d^\top y^* = 1$ makes y^* feasible, with $c^\top y^* = \lambda^+ / (1 - \lambda^+)$ attaining the bound. $\square$

Figure 1 shows the geometry in the shadow plane $\pi(x) = (d^\top x, c^\top x)$. Every $y \in T_P(v)$ projects into the tangent cone of the polygon $\pi(P)$ at $\pi(v)$; the constraint $d^\top y = 1$ is the vertical line at unit horizontal distance, and the program picks the lowest point of the cone on that line — the outgoing edge of the shadow boundary. The slope formula makes the convexity of the trade-off curve quantitative: since the map $\lambda \mapsto \lambda / (1 - \lambda)$ is increasing on $[0, 1)$ and consecutive vertices of the path depart at strictly increasing values λ^+ , the slopes strictly increase along the shadow path — each pivot leaves along a steeper edge than the last.

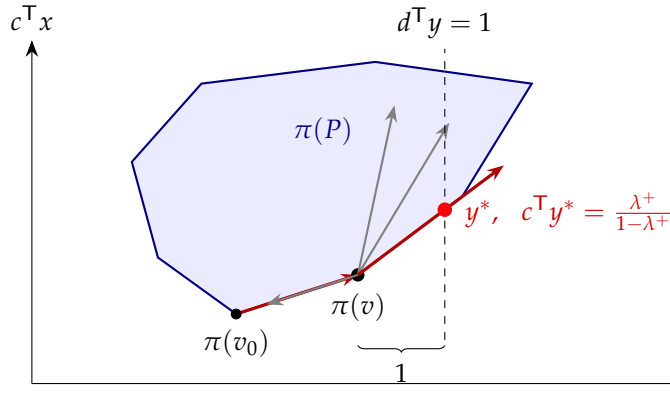


Figure 1: The tangent cone program (1) at a vertex v of the shadow path, drawn in the shadow plane $\pi(x) = (d^\top x, c^\top x)$: target d horizontal, cost c vertical. The directions of $T_P(v)$ project into the tangent cone of $\pi(P)$ at $\pi(v)$ (gray and red rays); among those of unit d -progress (the dashed line), the program picks the lowest — the outgoing edge of the trade-off curve, of slope $\lambda^+ / (1 - \lambda^+)$. No feasible direction dips below the shadow boundary, and slopes increase along the path.

With variable bounds, as in the flow LPs of Section 2, the tangent cone is just as explicit: at a vertex v of $P = \{x : Ax = b, 0 \leq x \leq u\}$ (or via the doubled form of Remark 1),

$$T_P(v) = \{y \in \ker(A) : y_j \geq 0 \ \forall j : v_j = 0, \ y_j \leq 0 \ \forall j : v_j = u_j\},$$

with free sign on the coordinates strictly between their bounds, and the tangent cone program becomes $\min c^\top y$ over this cone subject to $d^\top y = 1$. In Section 2 we will recognize this program, for the flow polytope with $d = \text{val}$, as a *shortest path problem* (Exercise 1).

2 Smoothed Analysis of the Successive Shortest Path Algorithm

Notation for this section: $G = (V, E)$ is a directed graph with $n := |V|$ nodes and $m := |E|$ arcs — the LP lives in dimension $|E|$, and its constraint matrix is the node-arc incidence matrix. The flow polytope below is a capacitated standard form, $Af = b, 0 \leq f \leq u$; via Remark 1 it is a standard-form program in the doubled variables $(f_a, \bar{f}_a = u_a - f_a)$, and the complement $\bar{f}_a$ of each arc is exactly its reverse residual

arc. We keep the residual-network language, which is the intuitive interpretation of this correspondence. To keep the notation unambiguous we assume that E contains at most one of each opposite pair ij, ji — without loss of generality, subdivide one of the two arcs — so that an ordered pair of nodes names at most one arc of E and at most one reversed arc.

2.1 Minimum-cost flow and the SSP algorithm

Let $G = (V, E)$ be a directed graph with source s , sink t , positive arc capacities $u \in \mathbb{R}_+^E$, and arc costs $c \in (0, 1)^E$ (a normalization). The feasible flows form the polytope

$$P := \left\{ f \in \mathbb{R}^E : \sum_{ij \in E} f_{ij} - \sum_{ji \in E} f_{ji} = 0 \quad \forall i \in V \setminus \{s, t\}, \quad \mathbf{0} \leq f \leq u \right\},$$

and the *value* of a flow is $\text{val}(f) := \sum_{sj \in E} f_{sj} - \sum_{js \in E} f_{js}$, the net flow leaving s . The *minimum-cost maximum-flow* problem asks for a flow of maximum value and, among those, of minimum cost $c^\top f$ — an LP over P .

Reduction from minimum-cost flow. The problem most often met in applications is *minimum-cost flow*: given demands $b \in \mathbb{R}^V$ with $\sum_v b_v = 0$, find f with $\mathbf{0} \leq f \leq u$ whose net outflow at each node v equals b_v , of minimum cost. This reduces to minimum-cost maximum flow as follows. Append a super-source s with a zero-cost arc sv of capacity b_v for every supply node ($b_v > 0$), and a super-sink t with a zero-cost arc vt of capacity $-b_v$ for every demand node ($b_v < 0$); see Figure 2. The flows of the enlarged network that saturate every auxiliary arc are exactly the feasible b -flows of the original network, with equal cost; and the maximum flows saturate the auxiliary arcs if and only if the original instance is feasible (the maximum value is at most $\sum_{v:b_v>0} b_v$, with equality precisely in that case). A minimum-cost maximum flow of the enlarged network therefore either certifies infeasibility or yields a minimum-cost b -flow. The auxiliary arcs carry deterministic cost 0, while the smoothed analysis below assumes random costs on *every* arc. The mismatch is harmless: in the feasible case every maximum flow saturates all auxiliary arcs, so assigning them arbitrary costs $c_a \in (0, 1)$ shifts the cost of every maximum flow by the same constant $\sum_{v:b_v>0} c_{sv} b_v + \sum_{v:b_v<0} c_{vt} |b_v|$, leaving the set of minimum-cost maximum flows unchanged. We may therefore smooth all costs, auxiliary ones included, and still recover a minimum-cost b -flow of the original instance.

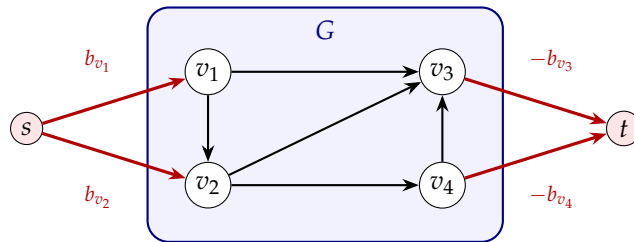


Figure 2: Reduction from minimum-cost flow to minimum-cost maximum flow. Supply nodes ($b_v > 0$) receive an arc from the super-source s , demand nodes ($b_v < 0$) an arc to the super-sink t (red); the auxiliary arcs have cost 0 and capacity $|b_v|$. The b -flows of G correspond to the flows of the enlarged network saturating all auxiliary arcs.

Circulations, and flow decomposition. Let $\mathbf{A}_G \in \mathbb{R}^{V \times E}$ denote the *node-arc incidence matrix* of G : the column of arc ij has entry $+1$ in row i and -1 in row j , so that $(\mathbf{A}_G w)_v$ is the net outflow of w at node v . The vectors $w \in \ker(\mathbf{A}_G)$ are the *circulations*: assignments of flow to arcs (negative entries meaning flow against the arc's direction) that are conserved at every node.

The structure theorem of network flow theory says that flows are superpositions of flows along paths and cycles. A crucial point of the statement is that the paths and cycles carry an *orientation relative to the arcs of G* : a simple path or cycle W in this lecture is a walk in G , visiting no node twice, each of whose edges traverses an arc $a \in E$ either *forwards* (in the direction of a) or *backwards* (against it); its signed indicator vector $\chi^W \in \{0, \pm 1\}^E$ records $\chi_a^W = +1$ for forward arcs, -1 for backward arcs, and 0 for unused ones. We call W *directed* when it traverses *all* of its arcs forwards.

THEOREM 2 (FLOW DECOMPOSITION) *Let $f \in P$ with $\text{val}(f) = \nu \geq 0$. Then*

$$f = \sum_i t_i \chi^{Q_i} + \sum_j t'_j \chi^{C_j}, \quad t, t' > 0, \quad \sum_i t_i = \nu,$$

where the Q_i are simple directed s - t paths and the C_j are simple directed cycles, at most $m + 1$ in total, all of whose arcs lie in $\text{supp}(f)$ and are traversed forwards.

More generally, every circulation $w \in \ker(\mathbf{A}_G) \setminus \{0\}$ is a positive combination $w = \sum_j t_j \chi^{C_j}$ of at most m simple cycles conforming to w : each C_j uses only arcs of $\text{supp}(w)$, traversing a forwards where $w_a > 0$ and backwards where $w_a < 0$.

We defer the proof to Section 3, where flow decomposition becomes the special case, for incidence matrices, of a decomposition principle valid for *every* constraint matrix (Exercise 3) — the form in which we will need it for general LPs in Section 4.

Flow decomposition is the workhorse of network flow theory. A typical use, central to Exercise 1 below: for flows $f, f' \in P$ of equal value, the difference $f' - f$ is a circulation, and the conforming cycles of its decomposition are cycles of the *residual network* defined next — traversing a forwards requires $f'_a > f_a$, hence $f_a < u_a$, and backwards requires $f'_a < f_a$, hence $f_a > 0$.

Write $\overleftarrow{E} := \{ji : ij \in E\}$ for the set of *reversed arcs*, $\overleftarrow{a} := ji$ for the reversal of $a = ij$, and extend the costs to $\overleftarrow{E}$ by $c_{\overleftarrow{a}} := -c_a$. (By our standing assumption, $E \cap \overleftarrow{E} = \emptyset$, so each ordered pair carries a well-defined cost.) For $f \in P$, the *residual network* $N[f]$ has node set V and arc set contained in $E \cup \overleftarrow{E}$: it contains $a \in E$ whenever $f_a < u_a$, and $\overleftarrow{a} \in \overleftarrow{E}$ whenever $f_a > 0$ — precisely the directions in which f has room to move. Augmenting along an s - t path in $N[f]$ increases the value while keeping feasibility. The classical SSP algorithm is:

Successive shortest paths. Start with $f = 0$. While $N[f]$ contains an s - t path, augment f as much as possible along a *minimum-cost* s - t path of $N[f]$. Return f .

We assume throughout that all minimum-cost paths arising are unique — true with probability 1 once costs are randomly perturbed. The fundamental (classical) facts about SSP are best stated through the *cost curve*

$$\Phi(\nu) := \min \{c^\top f : f \in P, \text{val}(f) = \nu\}, \quad \nu \in [0, \nu_{\max}],$$

a convex, piecewise linear function of the flow value (convexity: interpolate optimal flows, using that P is convex).

LEMMA 3 (SSP = WALKING THE COST CURVE) *Under the uniqueness assumption, the flows $f_0 = \mathbf{0}$, $f_1, \dots, f_k$ produced by the SSP algorithm are exactly the breakpoints of the cost curve Φ , in order. In particular, every f_i is a minimum-cost flow of its value, the final flow is a minimum-cost maximum flow, and the successive augmenting path costs*

$$s_i := \text{cost of the path used to go from } f_{i-1} \text{ to } f_i$$

are exactly the slopes of Φ , so that $0 < s_1 < s_2 < \dots < s_k < n - 1$.

This is a statement about the shadow vertex simplex method, whose correctness we have already justified: the curve Φ is the lower boundary of the two-dimensional shadow of P under $f \mapsto (\text{val}(f), c^\top f)$, i.e., for the pair (c, d) with $d := \text{val}$ — starting objective “minimize cost” (optimized uniquely by $f = \mathbf{0}$, as costs are positive), target objective $-\text{val}$, “maximize value”, exactly the orientation fixed in Section 1.1 — and the shadow walk visits its breakpoints in order, following at each vertex the edge selected by the tangent cone program (1), of strictly increasing slope (Theorem 1). All that remains is to check that *SSP implements this walk*: at each vertex f of the shadow path, the tangent cone program is solved by a maximal augmentation along the minimum-cost s - t path of $N[f]$ — one shortest-path computation per pivot. This is Exercise 1 below. The slope bounds hold since every augmenting path is simple with at most $n - 1$ arcs, each of cost in $(-1, 1)$, and the first slope, the cost of a cheapest s - t path in G itself, is positive.

Zadeh [Zad73] constructed instances on which the number of SSP iterations is exponential in the size of the network. As we now show, such instances do not survive even very small perturbations of the costs.

2.2 The smoothed bound

Perturbation model. Each cost c_a , $a \in E$, is an independent continuous random variable supported on $(0, 1)$ with probability density bounded by $\phi \geq 1$. Equivalently (*interval lemma*): $\Pr[c_a \in [\alpha, \beta]] \leq \phi \cdot (\beta - \alpha)$ for every interval $[\alpha, \beta] \subseteq [0, 1]$.

The canonical example: each c_a uniform on an adversarially chosen interval of length $1/\phi$ inside $[0, 1]$. As $\phi \rightarrow \infty$ the model degenerates to worst case, so ϕ measures the adversary’s precision; the theorem below says the adversary needs precision exponential in the instance size to force exponential behavior.

THEOREM 4 (SMOOTHED SSP, [BCM⁺15]) *For any network (G, u, s, t) with n nodes and m arcs, and random costs as above, the expected number of iterations of the SSP algorithm is $O(mn\phi)$.*

The proof strategy is a hallmark of smoothed analysis: we must find enough *independent* randomness so that it is not “used up” by conditioning on the algorithm’s history. The shadow vertex structure is what makes this possible: whether a flow appears on the cost curve depends only on the data, not on the path the algorithm took to it. The plan is to charge each iteration to an arc of G , and to bound, for each fixed arc a , the expected number of iterations charged to a by $O(n\phi)$, using only the randomness of c_a .

Arc sides. The charging scheme is indexed by the $2m$ elements of $E \cup \overleftarrow{E}$, which we call *arc sides*, each standing for one of the two bound constraints of its underlying arc:

$$a = ij \in E \longleftrightarrow \text{the lower bound } f_{ij} \geq 0, \quad \overleftarrow{a} = ji \in \overleftarrow{E} \longleftrightarrow \text{the upper bound } f_{ij} \leq u_{ij}.$$

(Recall E contains at most one of ij, ji , so an ordered pair determines its side unambiguously.) The correspondence pairs each side, viewed as a residual arc, with the bound that its traversal *releases*: pushing flow forward along ij raises f_{ij} off its lower bound, pushing backward along ji lowers f_{ij} off its upper bound. In the doubled standard form of Remark 1, the sides ij and ji are precisely the two nonnegative coordinates f_{ij} and $\bar{f}_{ij} = u_{ij} - f_{ij}$. For a side a , we write f_a at its bound to mean that the bound associated with a is tight ($f_{ij} = 0$ for $a = ij \in E$, $f_{ij} = u_{ij}$ for $a = ji \in \overleftarrow{E}$), and recall that c_a denotes the residual cost of a (c_{ij} , respectively $-c_{ij}$).

For $f \in P$, let $p(f)$ denote the cost of the minimum-cost s - t path in $N[f]$, and for a side $a \in E \cup \overleftarrow{E}$, let $p^{+a}(f)$ denote the minimum cost among simple s - t paths of $N[f]$ that use the residual arc a .

The charging step. Consider iteration i , moving from f_{i-1} along the shortest path R_i . The point f_{i-1} is a vertex of P , and the move releases *at least one* tight bound constraint: R_i uses some side a whose bound is tight at f_{i-1} — an arc $a = ij \in E$ of R_i with $f_{ij} = 0$, or a reversed arc $a = ji \in \overleftarrow{E}$ of R_i with $f_{ij} = u_{ij}$. (Otherwise the direction χ^{R_i} would lie in the kernel of every constraint tight at the vertex f_{i-1} , forcing $\chi^{R_i} = \mathbf{0}$.) Charge iteration i to one such side a (fixed arbitrarily); there are at most $2m$ sides. (In the doubled standard form, this is exactly charging each pivot to its entering coordinate.) Crucially, being charged to a means *the shortest path at f_{i-1} uses the residual arc a* :

$$p(f_{i-1}) = p^{+a}(f_{i-1}) = s_i. \quad (2)$$

The sandwich step. Fix a side $a \in E \cup \overleftarrow{E}$ and let $P_a := \{f \in P : f_a \text{ at its bound}\}$ denote the corresponding facet of P , with its own cost curve $\Phi_a(v) := \min\{c^\top f : f \in P_a, \text{val}(f) = v\}$, breakpoints $v_0^a, v_1^a, \dots$ and increasing slopes $s_1^a < s_2^a < \dots$, all lying in $(-(n-1), n-1)$. Augmenting within P_a must keep f_a at its bound, so it uses residual paths avoiding the underlying arc in *both* directions — avoiding a and $\overleftarrow{a}$ — and the slopes s_j^a are the costs of such a -avoiding shortest paths. Now, if iteration i is charged to a , then f_{i-1} lies on Φ_a : it is a minimum-cost flow of its value that is tight at a . In fact it is a *breakpoint* v_j^a of Φ_a , or an endpoint (absorbed by the additive 2 below): the curves are nested, $\Phi \leq \Phi_a$, and touch at $v^* := \text{val}(f_{i-1})$ (Figure 3), so their one-sided slopes pinch,

$$s_j^a = \Phi_a'^-(v^*) \leq \Phi'^-(v^*) = s_{i-1} < s_i = \Phi'^+(v^*) \leq \Phi_a'^+(v^*) = s_{j+1}^a,$$

and the strict slope break of Φ at v^* forces one of Φ_a as well. Combining with (2), and noting that the realized value $p^{+a}(v_j^a) = s_i$ is *positive* (all slopes of Φ are), we may clip the lower end at zero:

$$\max(s_j^a, 0) \leq p^{+a}(v_j^a) \leq s_{j+1}^a.$$

In particular, charges only ever land where Φ_a is *ascending* ($s_{j+1}^a > 0$); whether Φ_a has an initial descending stretch (possible for upper-bound sides) is immaterial. Distinct iterations charged to

a yield distinct breakpoints v_j^a , so

$$\#\{i : \text{iteration } i \text{ charged to } a\} \leq 2 + \sum_j \mathbf{1}[\max(s_j^a, 0) \leq p^{+a}(v_j^a) \leq s_{j+1}^a], \quad (3)$$

the +2 accounting for the extreme breakpoints of Φ_a .

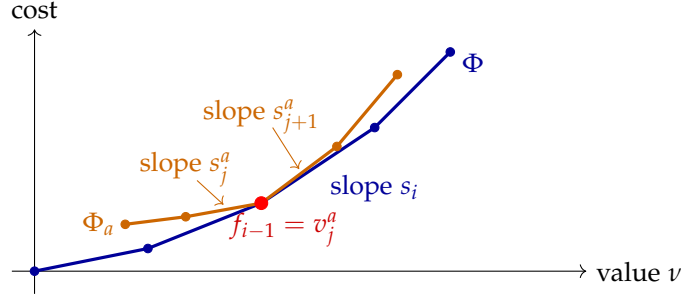


Figure 3: The sandwich step. The facet cost curve Φ_a (orange) lies above the full cost curve Φ (blue), both climbing with nonnegative, increasing slopes, and the two touch at the shared breakpoint f_{i-1} . Convexity of both curves forces the outgoing slope s_i of Φ to lie between the slopes $s_j^a \leq s_i \leq s_{j+1}^a$ of Φ_a around v_j^a .

The probabilistic step. Let $e \in E$ denote the arc underlying the charged side a (so that $c_a = \pm c_e$). We take expectations in (3), using only the randomness of c_e . The crucial observation is an *independence*: the facet curve Φ_a — its breakpoints v_j^a and its slopes s_j^a — does not depend on c_e . Indeed, on P_a the coordinate f_e is fixed at its bound, so changing c_e shifts the cost of every flow in P_a by the same constant, leaving all minimizers unchanged; and the slopes are costs of residual paths avoiding e in both directions. On the other hand, writing $a = pq$, every simple s – t path through the residual arc a decomposes into an s – p part, the arc a itself, and a q – t part, neither part meeting e again (by simplicity), so that

$$p^{+a}(v_j^a) = c_a + r_j = \pm c_e + r_j,$$

where r_j , the total cost of the cheapest such parts in $N[v_j^a]$, is again independent of c_e ; the sign is $+$ for $a \in E$ and $-$ for $a \in \overleftarrow{E}$. Hence, conditioned on all costs except c_e , the j -th indicator in (3) requires c_e to land in a *fixed* interval of length at most $\max(s_{j+1}^a, 0) - \max(s_j^a, 0)$ — empty whenever $s_{j+1}^a \leq 0$ — and the interval lemma bounds its probability by ϕ times that length. The sum over j telescopes:

$$\begin{aligned} \mathbb{E} \left[\sum_j \mathbf{1}[\max(s_j^a, 0) \leq p^{+a}(v_j^a) \leq s_{j+1}^a] \right] &\leq \phi \sum_j (\max(s_{j+1}^a, 0) - \max(s_j^a, 0)) \\ &\leq \phi \cdot \max_j \max(s_j^a, 0) \leq (n-1)\phi, \end{aligned}$$

since all slopes are less than $n-1$. Summing (3) over the at most $2m$ arc-sides:

$$\mathbb{E}[\#\text{iterations}] \leq 2m(2 + (n-1)\phi) = O(mn\phi). \quad \square$$

Let us summarize the key points of the analysis. The shadow vertex structure made the question *local*: whether a flow is visited is a property of the data alone, so we could condition

on everything except one coordinate of the randomness and still know the entire facet curve Φ_a . And the slopes provided a *bounded budget*: all the events charged to arc a occupy disjoint sub-intervals of a window of length $< n$, so their total probability is small no matter how many there are. Nearly matching lower bounds show the bound $O(mn\phi)$ is not far from the truth [BCM⁺15].

Exercise 1 (SSP implements the shadow vertex method) Let f be a vertex of P on the cost curve Φ (a minimum-cost flow of its value), not of maximum value. By the variable-bound description of the tangent cone (Section 1.2), the tangent cone program (1) at f for the pair (c, val) reads

$$\begin{aligned} \min \quad & c^\top y \\ \text{s.t.} \quad & (\mathbf{A}_G y)_v = 0 \quad \forall v \in V \setminus \{s, t\}, \quad \text{val}(y) = 1, \\ & y_a \geq 0 \quad \forall a \text{ with } f_a = 0, \quad y_a \leq 0 \quad \forall a \text{ with } f_a = u_a. \end{aligned}$$

- Show first that a flow is a minimum-cost flow of its value if and only if its residual network contains no negative-cost cycle. (Hint: Decompose $f' - f$ into conformal cycles for a competitor f' .)
- Show, using the conforming-cycle part of Theorem 2 (applied after appending an artificial arc ts of flow $\text{val}(y)$), that the paths and cycles in the decomposition of any feasible y travel only along residual arcs of $N[f]$.
- Show that the decomposition of a feasible y consists of s – t paths of $N[f]$ of total weight 1 together with residual cycles.
- Conclude with (a) that the optimal value of the program is the minimum cost of an s – t path in $N[f]$, attained at $y = \chi^Q$ for the cheapest path Q : the shadow vertex pivot at f is a maximal augmentation along the minimum-cost s – t residual path — an SSP iteration. Deduce Theorem 3.

Exercise 2 (Source and sink arcs are charged only once) The charging analysis is even simpler at the endpoints of the network.

- Show that over the run of SSP, the flow on every arc leaving s , and on every arc entering t , is monotonically nondecreasing.
- Conclude that in the proof of Theorem 4, each side of such an arc is charged at most once over the entire run — the forward side at most once, the reversed side never — so these sides need no randomness at all. Deduce an alternate proof that the deterministic zero-cost auxiliary arcs of the reduction from minimum-cost flow (Figure 2) are harmless: with only the original arcs smoothed, the expected number of iterations is still $O(mn\phi)$.

3 Circuit Decompositions & The Circuit Imbalance Measure

The analysis of Section 2 rested on two structural pillars of network flows: every flow decomposes into paths and cycles (Theorem 2), and the pieces of the decomposition have entries in $\{0, \pm 1\}$. This section develops the analogues of both facts for an *arbitrary* constraint matrix. They will be needed twice: in Section 4, where we carry out a more general version of the SSP analysis for LPs whose constraint matrix has bounded *circuit imbalance* κ ; and again in the interior point lectures, where circuits control the straight-line complexity of the central path (Lecture 4). Along the way, the flow decomposition theorem receives its promised proof (Exercise 3).

3.1 Circuits and conformal decomposition

The central objects are attached to a linear subspace.

DEFINITION 5 (CIRCUIT) Let $W \subseteq \mathbb{R}^n$ be a linear subspace. A circuit of W is a support-minimal nonzero vector $g \in W$: there is no nonzero $h \in W$ with $\text{supp}(h) \subsetneq \text{supp}(g)$. The circuits of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ are the circuits of $\ker(\mathbf{A})$ — equivalently, the minimal linear dependencies among the columns of $\mathbf{A}$.

The kernel of the constraint matrix is the setting of this lecture, but the notion depends only on the subspace, and applies just as well to, say, the row space $\text{im}(\mathbf{A}^\top)$.

Circuits play two roles: they are the pivot directions of the simplex method (Theorem 9 below), and they are the elementary building blocks into which every kernel vector decomposes *without cancellations* — the role played by simple cycles for circulations. The relevant notion is the following.

DEFINITION 6 (CONFORMAL VECTORS) For $g, w \in \mathbb{R}^n$, we say that g conforms to w if $\text{supp}(g) \subseteq \text{supp}(w)$ and $g_j w_j \geq 0$ for all j : every nonzero entry of g has the sign of the corresponding entry of w .

LEMMA 7 (CONFORMAL CIRCUIT DECOMPOSITION) Let $W \subseteq \mathbb{R}^n$ be a linear subspace. Every $w \in W \setminus \{0\}$ can be written as

$$w = g^1 + g^2 + \cdots + g^p, \quad p \leq \dim(W),$$

where each g^i is a circuit of W conforming to w . In particular, the sum is cancellation-free: $|w_j| = \sum_{i=1}^p |g_j^i|$ for every j .

PROOF: Consider the cone of vectors of W conforming to w ,

$$C_w := \{y \in W : y_j w_j \geq 0 \ \forall j \in \text{supp}(w), \ y_j = 0 \ \forall j \notin \text{supp}(w)\} \ni w.$$

C_w is a polyhedral cone, and it is *pointed*: if $\pm y \in C_w$, the sign constraints force $y = 0$. By Minkowski's theorem (Lecture 1), a pointed polyhedral cone is generated by its extreme rays: $C_w = \text{cone}(R(C_w))$.

Extreme rays of C_w are conforming circuits. Let g generate an extreme ray — a one-dimensional face of C_w , cut out by the constraints tight on it (Lecture 1). The tight constraints are the equations defining W together with $y_j = 0$ for every $j \notin \text{supp}(g)$, so they define the linear space

$$L := \{y \in W : y_j = 0 \ \forall j \notin \text{supp}(g)\} = \text{span}(g),$$

which is one-dimensional. Thus every $h \in W$ with $\text{supp}(h) \subseteq \text{supp}(g)$ lies in L , hence is collinear with g — no support can be strictly smaller, so g is a circuit, conforming to w by membership in C_w .

The decomposition. Since $w \in C_w = \text{cone}(R(C_w))$, w is a nonnegative combination of conforming circuits, and by Carathéodory's theorem (conic version, Lecture 1) we may keep only *linearly independent* ones, absorbing the positive coefficients into the circuits. They are linearly independent vectors of W , so $p \leq \dim(W)$. Cancellation-freeness is automatic: every g^i conforms to w , so on each coordinate all signs agree. $\square$

The flow decomposition theorem of Section 2 is exactly this statement for the node-arc incidence matrix, whose circuits are the simple cycles:

Exercise 3 (Flow decomposition is conformal decomposition) Deduce the flow decomposition theorem (Theorem 2) from Theorem 7 applied to the space of circulations $W = \ker(\mathbf{A}_G)$.

- (a) Show that the circuits of $\mathbf{A}_G$ are precisely the nonzero scalar multiples $t\chi^C$ of signed indicator vectors of simple cycles C .
- (b) Conclude the conforming-cycle statement: every circulation w is a positive combination of at most m simple cycles conforming to w — traversing each arc a forwards where $w_a > 0$ and backwards where $w_a < 0$.
- (c) Derive the path/cycle form for a flow $f \in P$ with $\text{val}(f) = v \geq 0$: append an artificial arc ts with $\tilde{f}_{ts} := v$, check that $\tilde{f}$ is a circulation, apply (b), and classify the cycles by whether they use ts .

3.2 The circuit imbalance measure

The second pillar of the SSP analysis — unit entries — is quantified by the following condition measure.

DEFINITION 8 (CIRCUIT IMBALANCE) *The circuit imbalance measure of a linear subspace $W \subseteq \mathbb{R}^n$ is*

$$\kappa = \kappa(W) := \max \left\{ \frac{|g_j|}{|g_i|} : g \text{ a circuit of } W, i, j \in \text{supp}(g) \right\},$$

the largest ratio between two entries of a circuit ($\kappa := 1$ if $W = \{\mathbf{0}\}$). For a matrix $\mathbf{A}$ we write $\kappa(\mathbf{A}) := \kappa(\ker(\mathbf{A}))$.

By the standard-form dictionary of Lecture 2, circuits are exactly the edge directions of standard-form polyhedra. In fact, they are exactly the directions the simplex method computes, and this gives a second, entirely concrete formulation of κ . Recall from Section 1.1 that when the entering index is $i \notin B$, the pivot direction is $w^{B,i}$ with $w_i^{B,i} = 1$, $w_B^{B,i} = -\mathbf{A}_B^{-1}\mathbf{A}_i$, and zeros elsewhere.

PROPOSITION 9 (CIRCUITS = PIVOT DIRECTIONS) *For every basis B and $i \notin B$, the pivot direction $w^{B,i}$ is a circuit, and every circuit of $\mathbf{A}$ is a scalar multiple of some $w^{B,i}$. Consequently,*

$$\kappa(\mathbf{A}) = \max \left\{ |(\mathbf{A}_B^{-1}\mathbf{A}_i)_j| : B \text{ basis}, i \in [n] \setminus B, j \in B \right\} :$$

κ is the largest entry, in absolute value, of any pivot direction ever computed by the simplex method, under the normalization $w_i = 1$ of the entering coordinate.

PROOF: $w^{B,i}$ is a circuit: it lies in $\ker(\mathbf{A})$ by construction, and if $h \in \ker(\mathbf{A})$ has $\text{supp}(h) \subsetneq \text{supp}(w^{B,i}) \subseteq B \cup \{i\}$, then either $h_i = 0$, so h is supported on B and $\mathbf{A}_B h_B = \mathbf{0}$ forces $h = \mathbf{0}$; or $h_i \neq 0$, and $h/h_i - w^{B,i}$ is supported on B , forcing $h/h_i = w^{B,i}$, contradicting the strict support inclusion.

Every circuit is of this form: let g be a circuit and pick any $i \in \text{supp}(g)$. The columns $\{\mathbf{A}_j : j \in \text{supp}(g) \setminus \{i\}\}$ are linearly independent (a dependency would be a kernel vector of strictly smaller support), and $\mathbf{A}_i$ lies in their span, so the rank of the columns $[n] \setminus \{i\}$ is still m and we may extend $\text{supp}(g) \setminus \{i\}$ to a basis $B \not\ni i$. Then $g/g_i - w^{B,i} \in \ker(\mathbf{A})$ is supported on B , hence zero.

The formula: by the above, the ratios $|g_j/g_i|$ over circuits g and $i, j \in \text{supp}(g)$ are exactly the numbers $|(\mathbf{A}_B^{-1} \mathbf{A}_i)_j|$; note every ratio and its reciprocal both occur (choose the pivot index i to be either endpoint), so the maximum is ≥ 1 and equals κ . $\square$

Two consequences we will use constantly: every circuit has $|\text{supp}(g)| \leq m+1$ (a pivot direction is supported on $B \cup \{i\}$), so

$$\|g\|_1 \leq (m+1) \|g\|_\infty \leq (m+1) \kappa |g_\ell| \quad \text{for every } \ell \in \text{supp}(g); \quad (4)$$

and κ depends on $\mathbf{A}$ only through $\ker(\mathbf{A})$, so it is invariant under row operations — though *not* under column scaling, which rescales circuit entries (minimizing κ over column rescalings defines the scaling-invariant variant κ^* ; see [DHN20] for its central role in LP theory).

For explicit bounds on κ , recall *Cramer's rule*: if $\mathbf{M} \in \mathbb{R}^{m \times m}$ is invertible, the unique solution of $\mathbf{M}x = b$ is

$$x_j = \frac{\det \mathbf{M}^{(j)}}{\det \mathbf{M}}, \quad j \in [m],$$

where $\mathbf{M}^{(j)}$ denotes $\mathbf{M}$ with its j -th column replaced by b . Applied to the system $\mathbf{A}_B x = \mathbf{A}_i$ — whose solution is $-w_B^{B,i}$ — it gives

$$(\mathbf{A}_B^{-1} \mathbf{A}_i)_j = \pm \frac{\det(\mathbf{A}_{B-j+i})}{\det(\mathbf{A}_B)}, \quad B-j+i := (B \setminus \{j\}) \cup \{i\}$$

(replacing column j of $\mathbf{A}_B$ by $\mathbf{A}_i$ changes the determinant only by the sign of a column permutation), and hence, by Theorem 9,

$$\kappa(\mathbf{A}) = \max \left\{ \frac{|\det(\mathbf{A}_{B-j+i})|}{|\det(\mathbf{A}_B)|} : B \text{ basis}, i \in [n] \setminus B, j \in B \right\} :$$

the circuit imbalance is the maximum ratio between the determinants of two bases differing in one column. In particular, for integral $\mathbf{A}$, $\kappa(\mathbf{A}) \leq \Delta_{\max}(\mathbf{A})$, the largest absolute value of an $m \times m$ subdeterminant of $\mathbf{A}$.

The determinant formula reveals a symmetry that the kernel definition hides:

Exercise 4 (Circuit imbalance is self-dual) Prove that for every linear subspace $W \subseteq \mathbb{R}^n$,

$$\kappa(W) = \kappa(W^\perp).$$

(Hint: Choose coordinates so that $W = \ker(\mathbf{A})$ for $\mathbf{A} = [\mathbf{I}_m \mid \mathbf{M}]$. Give a representation of $W^\perp$ as the kernel of a matrix depending on $\mathbf{M}$, and compare its subdeterminants with those of $\mathbf{A}$.)

Examples. If $\mathbf{A}$ is totally unimodular — every subdeterminant in $\{0, \pm 1\}$, e.g. a node-arc incidence matrix, as in Section 2 — then $\kappa = 1$: indeed, circuits of incidence matrices are simple cycles with ± 1 entries, as verified in Exercise 3. The measure degrades gracefully under the operations we will need, which we state, like the notion itself, for an arbitrary subspace:

LEMMA 10 (STABILITY OF κ) *Let $W \subseteq \mathbb{R}^n$ be a linear subspace, $i \in [n]$, and $\sigma \in \{\pm 1\}$. The circuit imbalance does not increase under any of the following operations on W .*

- (i) Deletion: $W \setminus i := \{y_{-i} : y \in W, y_i = 0\}$ — restrict to $\{y_i = 0\}$ and drop the coordinate.

(ii) Contraction: $W/i := \{y_{-i} : y \in W\}$ — project the coordinate away.

(iii) Duplication up to sign: $W^{i,\sigma} := \{(y, t) \in \mathbb{R}^{n+1} : y + \sigma t e_i \in W\}$ — a new coordinate playing the role of σy_i .

On kernels, these are matrix operations: deleting column i of $\mathbf{A}$ performs (i) on $\ker(\mathbf{A})$; applying row operations to $\mathbf{A}$ until column i equals some unit vector e_j — row operations do not change the kernel; if $\mathbf{A}_i = \mathbf{0}$, simply delete the column — and then deleting row j and column i performs (ii); and appending the column $\sigma \mathbf{A}_i$ performs (iii).

PROOF: (i) A circuit of $W \setminus i$, with a zero i -th entry re-inserted, is a vector of W , and it is support-minimal there: any $h \in W$ of strictly smaller support also has $h_i = 0$, hence descends to $W \setminus i$, contradicting minimality. Its entries, and so its ratios, are unchanged.

(ii) Let g' be a circuit of W/i , and let $\tilde{g} \in W$ be a lift ($\tilde{g}_{-i} = g'$) of minimal support, so $\text{supp}(\tilde{g}) \subseteq \text{supp}(g') \cup \{i\}$. We claim $\tilde{g}$ is a circuit of W ; its entries include those of g' , so its ratios are at most $\kappa(W)$, as needed. Suppose $h \in W$ is nonzero with $\text{supp}(h) \subsetneq \text{supp}(\tilde{g})$; then $h_{-i} \in W/i$ has support inside $\text{supp}(g')$. If $h_{-i} = 0$, then $h = t e_i$ with $t \neq 0$, and $\tilde{g} - (\tilde{g}_i/t) h$ is a lift of g' of smaller support — impossible. Otherwise minimality of g' forces $h_{-i} = c g'$ with $c \neq 0$, and $\tilde{g} - c^{-1} h$ is a multiple of e_i : if nonzero, then $e_i \in W$ and $\tilde{g} - \tilde{g}_i e_i$ is a lift of smaller support — impossible; if zero, then $\tilde{g} = c^{-1} h$, contradicting the strict inclusion.

(iii) The pair vector $p := (-\sigma e_i, 1) \in W^{i,\sigma}$ has two entries of equal magnitude: ratio 1. A circuit of $W^{i,\sigma}$ using both coordinates i and $n+1$ must be a multiple of p : otherwise, subtracting the multiple of p that cancels its i -th entry strictly shrinks the support. Every other circuit uses at most one of the two coordinates and corresponds, after identifying coordinate $n+1$ with σy_i , to a circuit of W with the same entry magnitudes. $\square$

With the decomposition and the imbalance measure in hand, we now extend the smoothed analysis from flows to general LPs.

4 Smoothed Analysis via Circuit Imbalances

(We return to the matrix conventions of Section 1.1: $\mathbf{A} \in \mathbb{R}^{m \times n}$ with m rows and $n \geq m$ columns, full row rank.)

Let us examine what the smoothed analysis of Section 2 actually used. Three of its ingredients were generic: the charging of pivots to entering coordinates, the independence of the facet curve from the charged cost, and the nested-curves sandwich. Two were flow-specific: every augmenting structure had entries in $\{0, \pm 1\}$, so the charged cost entered path costs with unit coefficient, and path costs were bounded by $n-1$, so slopes lived in a *bounded window*. In this section we extend the analysis to general LPs

$$\min c^\top x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0},$$

with smoothed costs. The unit coefficients will be replaced by the circuit imbalance measure κ of Section 3.2 — augmenting structures are now the circuits of $\mathbf{A}$, by way of the conformal decomposition of Section 3.1 — and the bounded window will be restored not by an assumption but by an *algorithm design*: instead of one shadow interpolation, we insert the constraints one at a time, interpolating each new right-hand side along a fresh auxiliary coordinate.

Standing assumptions: the costs c_j are independent random variables supported on $(0, 1]$ with density at most $\phi \geq 1$ (in particular $c > \mathbf{0}$ almost surely); after row operations and a permutation of columns, $\mathbf{A}_{:, [m]} = \mathbf{I}_m$, i.e., the first m columns form an identity (row operations change neither the feasible region nor, as we note below, κ); and each truncated system $(\mathbf{A}_{[k], [n]}, b_{[k]}), k \in [m]$, is non-degenerate. No sign assumption on b is needed.

4.1 The algorithm: inserting one constraint at a time

The algorithm builds up the constraint set row by row. For $k \in [m]$, consider the *phase- k trade-off curve*

$$\Phi_k(s) := \min \left\{ c^\top x : \mathbf{A}_{[k-1], [n]} x = b_{[k-1]}, \quad a_k^\top x = s, \quad x \geq \mathbf{0} \right\}, \quad s \in I_k,$$

where $a_k^\top$ is the k -th row and I_k is the (interval) set of attainable values of $a_k^\top x$; for $k = 1$ there are no equality constraints besides $a_1^\top x = s$. As in Section 2, Φ_k is convex and piecewise linear, its breakpoints are (generically) vertices, and walking it is a shadow path: the parametric objectives $c - \lambda s$ (for $\lambda \geq 0$, moving right) or $c + \lambda s$ (moving left) trace it out, one pivot per breakpoint, exactly as in the shadow vertex rule of Section 1.1.

Constraint-insertion shadow simplex. Set $x^0 := \mathbf{0}$. For $k = 1, \dots, m$: starting from $(x^{k-1}, \sigma_k := a_k^\top x^{k-1})$, walk the curve Φ_k monotonically from $s = \sigma_k$ to $s = b_k$; if $b_k \notin I_k$, stop and report *infeasible*. Otherwise let x^k be the point reached at $s = b_k$. Return x^m .

Two small lemmas show how the phases fit together. Recall $\mathbf{A}_{:, [m]} = \mathbf{I}_m$: the variable x_j , for $j \in \{k, \dots, m\}$, appears in row j and in *no* earlier row.

LEMMA 11 (IDLE VARIABLES REST) *Every point on the curve Φ_k has $x_{k+1} = \dots = x_m = 0$. In particular $x_j^k = 0$ for $j > k$.*

PROOF: For $j > k$, the variable x_j appears in no constraint of phase k except $x_j \geq 0$, and $c_j > 0$: lowering it to 0 preserves feasibility and strictly lowers the cost. $\square$

LEMMA 12 (CHAINING) *For each k , the point (x^{k-1}, σ_k) lies on Φ_k and is its global minimum.*

PROOF: Since s is unconstrained, minimizing $c^\top x$ over the phase- k region is minimizing over $\{x \geq \mathbf{0} : \mathbf{A}_{[k-1], [n]} x = b_{[k-1]}\}$ — which is precisely the problem whose optimum is x^{k-1} (for $k = 1$: the orthant, minimized at $\mathbf{0}$ as $c > \mathbf{0}$). Its row- k value is σ_k by definition. $\square$

Thus each phase starts at the global minimum of its curve and walks to the prescribed level $s = b_k$, on whichever side of the minimum b_k lies (Figure 4). If some phase finds $b_k \notin I_k$, then no $x \geq \mathbf{0}$ satisfies the first k equalities, and the LP is infeasible; unboundedness never occurs since $c > \mathbf{0}$; and after phase m , the output x^m minimizes c subject to all m constraints. The algorithm is thus a complete LP solver for smoothed c — no separate Phase 0 needed.

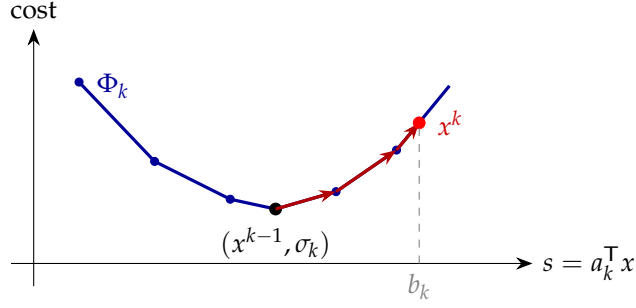


Figure 4: Phase k of the constraint-insertion algorithm. By Theorem 12, the endpoint of phase $k - 1$ sits at the bottom of the phase- k curve; the walk climbs to the prescribed level $s = b_k$ (mirrored if $b_k < \sigma_k$). Every slope encountered is the ratio $c^T w / w_s$ of a circuit, hence at most $(k + 1)\kappa$ in absolute value.

4.2 The analysis

The phase- k constraint matrix, in the variables (x, s) , is

$$\mathbf{M}_k := [\mathbf{A}_{[k],[n]} \mid -e_k],$$

since the system reads $\mathbf{A}_{[k-1],[n]} x = b_{[k-1]}, a_k^T x - s = 0$. The central point of the design is that *the auxiliary column is a copy of a column already present* (column k of $\mathbf{A}_{[k],[n]}$ is e_k , as $\mathbf{A}_{\cdot,[m]} = \mathbf{I}_m$), and the interpolation functional is the *coordinate* s . Cancellations in “ $d^T w$ ” are therefore impossible — the quantity is an entry of the circuit — and Theorem 10 keeps the condition measure under control:

LEMMA 13 (PHASE SYSTEMS INHERIT κ) $\kappa(\mathbf{M}_k) \leq \kappa(\mathbf{A})$ for every $k \in [m]$, and every circuit of $\mathbf{M}_k$ has support of size at most $k + 1$.

PROOF: We apply Theorem 10 to the subspace $W := \ker(\mathbf{A})$. Contracting (projecting away) the coordinates $x_m, \dots, x_{k+1}$ in turn does not increase κ ; since column j is the unit column e_j for each $j > k$, this is the matrix contraction of Theorem 10(ii), and the result is $\ker(\mathbf{A}_{[k],[n] \setminus \{k+1, \dots, m\}})$. Restoring the columns $k + 1, \dots, m$ — zero columns, as they vanish on the first k rows — turns the kernel into its direct sum with the free coordinates $x_{k+1}, \dots, x_m$, whose only new circuits are the singletons e_j , of ratio 1: this gives $\ker(\mathbf{A}_{[k],[n]})$. Finally, appending the column $-e_k$, a negated duplicate of column k , is duplication up to sign (Theorem 10(iii)) and yields $\ker(\mathbf{M}_k)$. The support bound holds since $\mathbf{M}_k$ has rank k . $\square$

THEOREM 14 (SMOOTHED CONSTRAINT INSERTION) *Under the standing assumptions, phase k performs at most $n(2 + (k + 1)\kappa^2\phi)$ pivots in expectation, and hence the expected total over all m phases is*

$$\mathbb{E}[\#pivots] \leq 2nm + 2nm^2\kappa^2\phi = O(nm^2\kappa^2\phi).$$

PROOF: Fix a phase k and suppose $b_k > \sigma_k$, so the walk climbs the right branch (the left branch is the mirror image). The proof of Theorem 4 applies word for word; we record the dictionary.

Slopes and the window. Each edge of the walk has direction a circuit (w, w_s) of $\mathbf{M}_k$ with $s \in \text{supp}$, and slope $\sigma = c^T w / w_s$. By Theorem 13 and (4), $|\sigma| \leq \|c\|_\infty \|w\|_1 / |w_s| \leq (k + 1)\kappa$. The

same holds on every facet curve below. All slopes on the walked branch lie in $[0, (k+1)\kappa]$, and they strictly increase.

Charging. The only inequality constraints are $x \geq 0$, so each pivot moves off at least one tight bound $x_a = 0$; charge the pivot to its *entering coordinate* $a \in [n]$, as in Section 2.

Sandwich. Fix a and let $\Phi_{k,a}$ be the curve of the facet $\{x_a = 0\}$ — the trade-off curve of the phase- k system with column a deleted, which involves neither the variable x_a nor the cost c_a . If a pivot at breakpoint v is charged to a , then v lies on both curves, $\Phi_k \leq \Phi_{k,a}$, the curves touch at v , and by convexity the outgoing slope of Φ_k at v — which equals the best entering- a slope, $\min\{c^\top w/w_s\}$ over a -entering tangent circuits (w, w_s) at v with $w_s > 0$ (rightward moves) — lies between the two slopes of $\Phi_{k,a}$ adjacent to v , exactly as in Figure 3. Distinct charged pivots yield distinct breakpoints of $\Phi_{k,a}$.

Sweep. Condition on $(c_j)_{j \neq a}$: the curve $\Phi_{k,a}$, its breakpoints and its slopes are determined. At each of its breakpoints, the best entering- a slope is a minimum of functions affine in c_a with derivatives $w_a/w_s \in [1/\kappa, \kappa]$ — ratios of two *positive* entries of a single circuit ($w_a > 0$ as a enters, $w_s > 0$ on the right branch), by Theorem 13 — hence a piecewise affine increasing function of c_a with derivative at least $1/\kappa$. The sandwich event confines c_a to a single interval of length at most κ times the adjacent slope gap of $\Phi_{k,a}$, so by the interval lemma its probability is at most $\phi\kappa \cdot (\text{gap})$. Summing over the breakpoints of $\Phi_{k,a}$, the gaps telescope — clipped at zero exactly as in Theorem 4, since a charged pivot realizes the nonnegative outgoing slope of Φ_k — to at most the slope window $(k+1)\kappa$, giving expected charge at most $2 + (k+1)\kappa^2\phi$ per coordinate a (the $+2$ for the extreme breakpoints). Summing over $a \in [n]$ proves the phase bound, and $\sum_{k=1}^m (k+1) \leq m(m+3)/2 \leq 2m^2$ the total. $\square$

Discussion. For node-arc incidence matrices ($\kappa = 1$) the guarantee is a factor $\sim m$ short of the SSP bound $O(nm\phi)$ — the price of running m separate interpolations where the flow structure supports a single one (there, “insert the conservation constraints one at a time” is subsumed by pushing flow value). Whether the phases can be amortized to remove this factor is, to our knowledge, open. The design principle deserves emphasis: the difficulty in smoothed analyses of general shadow paths is that the fixed interpolation objective d can pair degenerately with circuits ($d^\top w \approx 0$ while w is large), which no bound in $\kappa(\mathbf{A})$ alone can exclude; making d a *coordinate of the system* — at the cost of one auxiliary variable per phase — removes the problem structurally. An alternative route, which randomizes the interpolation objectives instead and applies even to worst-case c , was developed by Brunsch and Röglin [BR13] in the language of the δ -distance property; see also [DH16] for sharper results in this parameter regime.

Looking ahead. The two analyses of this lecture explain the practical success of the simplex method in probabilistic terms: perturbing the objective collapses the shadow path to (expected) polynomial length, at a rate governed by the circuit structure of the constraint matrix. What randomness cannot provide is an unconditional guarantee. In the next lecture we change strategy entirely: instead of walking the boundary from vertex to vertex, we will travel through the *interior* of the polyhedron along the *central path* — the realm of interior point methods, where polynomial-time guarantees hold unconditionally — and study how the geometry of this path governs their iteration complexity. For further reading on this lecture’s material, see the chapter [DH21] that our smoothed analysis follows, [BCM⁺15] for the original SSP analysis and nearly

matching lower bounds, and [DHN^V20] for the central role of the circuit imbalance measure in linear programming.

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